

Mark Rober's CrunchLabs My Domino Robot Broke a World Record

Time	Subtitle	Human Translation
0s	[Mark] This is Dom.	[Mark] This is Dom.
1s	He's a friendly little robot that's supergood at only one thing,	He's a friendly little robot that's supergood at only one thing,
4s	setting up a butt-ton of dominoes really, really fast.	setting up a butt-ton of dominoes really, really fast.
8s	It's taken five years to get to this point, but according to our tests,	It's taken five years to get to this point, but according to our tests,
11s	we have hopes he might be more than ten times faster	we have hopes he might be more than ten times faster
15s	than the fastest human.	than the fastest human.
16s	We're gonna put him to the test,	We're gonna put him to the test,
17s	head-to-head against the world's foremost domino expert,	head-to-head against the world's foremost domino expert,
20s	who unfortunately is a mere human.	who unfortunately is a mere human.
23s	Then we'll attempt a Guinness World Record,	Then we'll attempt a Guinness World Record,
25s	and he's gonna try and fill this entire warehouse	and he's gonna try and fill this entire warehouse
27s	with a mural of 100,000 dominoes.	with a mural of 100,000 dominoes.
30s	And for context, setting up a domino mural that size	And for context, setting up a domino mural that size
33s	would take a team of seven skilled builders one full week.	would take a team of seven skilled builders one full week.
36s	Dom here is gonna attempt to do it by himself in 24 hours.	Dom here is gonna attempt to do it by himself in 24 hours.
40s	I know what you're thinking. "Big whoop, Mark."	I know what you're thinking. "Big whoop, Mark."
42s	Because as a kid, you might've had a toy like this,	Because as a kid, you might've had a toy like this,
45s	or even this custom robot KiwiCo made me that can drop 20 dominoes in a row.	or even this custom robot KiwiCo made me that can drop 20 dominoes in a row.
49s	So isn't it overkill to utilize an autonomous robot,	So isn't it overkill to utilize an autonomous robot,
52s	a half mile of Hot Wheels tracks,	a half mile of Hot Wheels tracks,
54s	and a high-speed robotic arm?	and a high-speed robotic arm?
56s	I get it, because our designs looked like this,	I get it, because our designs looked like this,
58s	where we'd just hack a little Roomba,	where we'd just hack a little Roomba,
1:00	when we first tried to tackle this problem exactly five years ago.	when we first tried to tackle this problem exactly five years ago.

1:04	But then pesky details with scale arise.	But then pesky details with scale arise.
1:07	Like, for starters, just how much 100,000 dominoes is.	Like, for starters, just how much 100,000 dominoes is.
1:11	So the robot will need to come and reload at least a thousand times,	So the robot will need to come and reload at least a thousand times,
1:15	but how exactly do you reload it?	but how exactly do you reload it?
1:18	And how would it know exactly where to drop each domino,	And how would it know exactly where to drop each domino,
1:21	and what color it should be?	and what color it should be?
1:22	And how do you make it so reliable that it doesn't screw up once	And how do you make it so reliable that it doesn't screw up once
1:26	in 100,000 drops of a domino?	in 100,000 drops of a domino?
1:28	A system to reliably tackle issues at scales like this	A system to reliably tackle issues at scales like this
1:32	is going to be inherently super complex.	is going to be inherently super complex.
1:34	Like, ten times more complex than my automatic bull's-eye dartboard,	Like, ten times more complex than my automatic bull's-eye dartboard,
1:38	which to this point was probably the most complicated build on my channel.	which to this point was probably the most complicated build on my channel.
1:41	After failing off and on for three years,	After failing off and on for three years,
1:44	I was doing a Q&A at Maker Faire	I was doing a Q&A at Maker Faire
1:45	and told the crowd if anyone wanted to help me with a brutal challenge,	and told the crowd if anyone wanted to help me with a brutal challenge,
1:49	to hit me up afterwards.	to hit me up afterwards.
1:50	To my absolute delight,	To my absolute delight,
1:51	two freshmen from Stanford and a software engineer from the Bay Area	two freshmen from Stanford and a software engineer from the Bay Area
1:55	took me up on the challenge.	took me up on the challenge.
1:56	So I hired them, and two years later, here we were.	So I hired them, and two years later, here we were.
1:59	Before I show you how it works,	Before I show you how it works,
2:01	I wanted to put him in a head-to-head competition	I wanted to put him in a head-to-head competition
2:03	to see how good he really was.	to see how good he really was.
2:05	For that, we need the undisputed heavyweight champion on YouTube	For that, we need the undisputed heavyweight champion on YouTube
2:08	for all things dominoes, Lily Hevesh.	for all things dominoes, Lily Hevesh.
2:11	Lily, you are known on YouTube as the Queen of Dominoes, right?	Lily, you are known on YouTube as the Queen of Dominoes, right?
2:16	People do call me the Domino Queen.	People do call me the Domino Queen.
2:18	[Mark] And for very good reason.	[Mark] And for very good reason.
2:19	With over 1.2 billion views on her channel,	With over 1.2 billion views on her channel,
2:21	you've almost certainly seen her incredible creations	you've almost certainly seen her incredible creations

2:24	over her 12 years on the platform.	over her 12 years on the platform.
2:26	She's so good, she's even got her own line of dominoes you can buy in stores.	She's so good, she's even got her own line of dominoes you can buy in stores.
2:30	Here's the challenge, Lily.	Here's the challenge, Lily.
2:32	I want you to set up 300 dominoes right now.	I want you to set up 300 dominoes right now.
2:35	Let's see how long it takes a human to do that.	Let's see how long it takes a human to do that.
2:37	-Okay. -[Mark] Okay?	-Okay. -[Mark] Okay?
2:38	Clock starts... now.	Clock starts... now.
2:43	I gotta do two hands.	I gotta do two hands.
2:46	If you need me, Lily, just sitting here reading my newspaper.	If you need me, Lily, just sitting here reading my newspaper.
2:50	For you kids at home, a newspaper is like a boring iPad made from trees.	For you kids at home, a newspaper is like a boring iPad made from trees.
2:56	[chuckles] That's actually pretty fast.	[chuckles] That's actually pretty fast.
3:01	-Oh no! The middle of the field. -[Mark] Disaster.	-Oh no! The middle of the field. -[Mark] Disaster.
3:04	The foibles of being a human.	The foibles of being a human.
3:07	-Okay, Lily, are you ready for this? -No.	-Okay, Lily, are you ready for this? -No.
3:10	I've seen enough.	I've seen enough.
3:12	I don't know how to whistle.	I don't know how to whistle.
3:14	This	This
3:16	is the Dominator.	is the Dominator.
3:18	-[Mark laughs] -Oh my God.	-[Mark laughs] -Oh my God.
3:20	Wow.	Wow.
3:22	I love the eyes.	I love the eyes.
3:23	[Mark] A good set of googly eyes will go a long ways.	[Mark] A good set of googly eyes will go a long ways.
3:27	Okay, boy,	Okay, boy,
3:28	let 'em down.	let 'em down.
3:30	I suck at dominoes, Lily,	I suck at dominoes, Lily,
3:32	but I'm good at engineering, which means...	but I'm good at engineering, which means...
3:35	-I'm actually really good at dominoes. -No!	-I'm actually really good at dominoes. -No!
3:38	[Mark laughs] We even showed off.	[Mark laughs] We even showed off.
3:40	The Dominator double-stacked the middle row for you.	The Dominator double-stacked the middle row for you.
3:43	That is impressive.	That is impressive.
3:45	That is seriously impressive.	That is seriously impressive.
3:47	[Mark] This technique of laying down a tile of 300 dominoes at once	[Mark] This technique of laying down a tile of 300 dominoes at once

3:50	was the key that helped us crack the scale and reliability issues.	was the key that helped us crack the scale and reliability issues.
3:53	It's probably my favorite part about the Dominator,	It's probably my favorite part about the Dominator,
3:56	besides his wiggly tail, of course.	besides his wiggly tail, of course.
3:58	Here's Josh to explain more.	Here's Josh to explain more.
3:59	We have these 3D printed funnels	We have these 3D printed funnels
4:01	which allow us to drop all 300 dominoes at once.	which allow us to drop all 300 dominoes at once.
4:03	And the mechanism that drops those into the top of this tray,	And the mechanism that drops those into the top of this tray,
4:06	we'll show that in a second.	we'll show that in a second.
4:08	[Josh] And if we remove these funnels,	[Josh] And if we remove these funnels,
4:10	we see that the dominoes are precisely placed in their locations.	we see that the dominoes are precisely placed in their locations.
4:13	This allows us to place 300 dominoes where we want 'em all at once.	This allows us to place 300 dominoes where we want 'em all at once.
4:17	[Mark] They don't fall out because of what we call the Connect Four mechanism.	[Mark] They don't fall out because of what we call the Connect Four mechanism.
4:21	But if we trigger the servo, it slides that Connect Four tray over	But if we trigger the servo, it slides that Connect Four tray over
4:25	and drops them all at once,	and drops them all at once,
4:26	making for some gratuitous, beautiful slo-mo shots.	making for some gratuitous, beautiful slo-mo shots.
4:42	While this looks really cool,	While this looks really cool,
4:44	dropping from that height wouldn't be effective.	dropping from that height wouldn't be effective.
4:46	So we lowered the whole platform down on this ball screw,	So we lowered the whole platform down on this ball screw,
4:49	then set them right on the floor,	then set them right on the floor,
4:51	which turned out to be incredibly reliable.	which turned out to be incredibly reliable.
4:53	So Dom took the W in round one, but now it's time for the endurance round.	So Dom took the W in round one, but now it's time for the endurance round.
4:57	Okay, first one to 10,000 dominoes wins.	Okay, first one to 10,000 dominoes wins.
5:01	Lily, you ready?	Lily, you ready?
5:02	I'm ready.	I'm ready.
5:03	Dominator, you ready?	Dominator, you ready?
5:07	Start.	Start.
5:12	[Lily] Oh my God. Wow.	[Lily] Oh my God. Wow.
5:14	-I already failed. -[Mark] That's great.	-I already failed. -[Mark] That's great.
5:23	To Lily's credit, she was actually insanely fast,	To Lily's credit, she was actually insanely fast,
5:26	just not,	just not,
5:28	you know, robot fast.	you know, robot fast.

5:30	So unfair.	So unfair.
5:32	So unfair.	So unfair.
5:33	[Mark] On top, she had to handle a large distraction	[Mark] On top, she had to handle a large distraction
5:36	with the emotional maturity of a nine-year-old. [chuckles]	with the emotional maturity of a nine-year-old. [chuckles]
5:39	Uh, yep, I think I'm doing pretty good.	Uh, yep, I think I'm doing pretty good.
5:46	You're doing very good for a human, I'll give you that.	You're doing very good for a human, I'll give you that.
5:50	Oh no.	Oh no.
5:57	-No what? No what, Lily? -[Lily] Get that out of here. No. Get out.	-No what? No what, Lily? -[Lily] Get that out of here. No. Get out.
6:01	Agh!	Agh!
6:04	That feels so much faster than it probably looks.	That feels so much faster than it probably looks.
6:07	-How's Lily doing? -Tired.	-How's Lily doing? -Tired.
6:09	[Mark] So while Lily's output slowed down considerably,	[Mark] So while Lily's output slowed down considerably,
6:12	Dom's did not. I should mention, we sometimes call him Dom,	Dom's did not. I should mention, we sometimes call him Dom,
6:15	but that's short for his legal birth name, which, as noted before, is Dominator.	but that's short for his legal birth name, which, as noted before, is Dominator.
6:20	'Cause, you know, like domino, dominator.	'Cause, you know, like domino, dominator.
6:23	-Whatever. It all works. -I'm gonna head to the bathroom.	-Whatever. It all works. -I'm gonna head to the bathroom.
6:26	Oh! Lily has to go to the bathroom.	Oh! Lily has to go to the bathroom.
6:30	I'm human, right?	I'm human, right?
6:31	Dominator, do you have to go to the bathroom?	Dominator, do you have to go to the bathroom?
6:34	Didn't think so.	Didn't think so.
6:39	Nine thousand perfectly laid dominoes in a little over two hours.	Nine thousand perfectly laid dominoes in a little over two hours.
6:43	Actually, 9,001.	Actually, 9,001.
6:45	We got a bonus domino out there in the middle of this field.	We got a bonus domino out there in the middle of this field.
6:49	May or may not have been on purpose, but now we can say it's over 9,000.	May or may not have been on purpose, but now we can say it's over 9,000.
6:53	It was a clear result.	It was a clear result.
6:54	And Lily, of course, was gracious in her defeat to the machine overlords.	And Lily, of course, was gracious in her defeat to the machine overlords.
6:58	All that was left to do now was to knock 'em down.	All that was left to do now was to knock 'em down.
7:05	-[chuckles] Yes! That was satisfying. - Whoo!	-[chuckles] Yes! That was satisfying. - Whoo!

7:08	[Mark] We went head-to-head with the world's foremost domino expert	[Mark] We went head-to-head with the world's foremost domino expert
7:11	and came out on top.	and came out on top.
7:12	Now time for the real challenge.	Now time for the real challenge.
7:14	To attempt a Guinness World Record	To attempt a Guinness World Record
7:16	and see how quickly we can fill this entire room	and see how quickly we can fill this entire room
7:19	with a mural of over 100,000 dominoes.	with a mural of over 100,000 dominoes.
7:22	All right, buddy, you got the game plan. Now just get out there and execute it.	All right, buddy, you got the game plan. Now just get out there and execute it.
7:27	Godspeed, little fella. Godspeed.	Godspeed, little fella. Godspeed.
7:36	Three hundred dominoes down,	Three hundred dominoes down,
7:38	102,000 left to go.	102,000 left to go.
7:41	♪ Enchanté ♪	♪ Enchanté ♪
7:52	♪ Enchanté ♪	♪ Enchanté ♪
8:04	[Mark] So we're about six hours in, but more importantly,	[Mark] So we're about six hours in, but more importantly,
8:07	we just got our first Goomba.	we just got our first Goomba.
8:10	It's beautiful!	It's beautiful!
8:11	Back to work, Dom.	Back to work, Dom.
8:15	The best part about working really hard for a few years	The best part about working really hard for a few years
8:17	is that the last 24 hours are super relaxing.	is that the last 24 hours are super relaxing.
8:23	And with that tile placed, we are officially halfway done	And with that tile placed, we are officially halfway done
8:28	with over 50,000 dominoes put down	with over 50,000 dominoes put down
8:31	in just a little bit over 12 hours.	in just a little bit over 12 hours.
8:33	And Dom... Whoa, easy, boy.	And Dom... Whoa, easy, boy.
8:36	Just showing no signs of slowing down.	Just showing no signs of slowing down.
8:38	This is a good point to explain the robotic system.	This is a good point to explain the robotic system.
8:41	It all starts over here with the loading station	It all starts over here with the loading station
8:43	that Josh and John Luke spent a lot of time optimizing.	that Josh and John Luke spent a lot of time optimizing.
8:46	Dominoes are loaded by color on conveyor belts,	Dominoes are loaded by color on conveyor belts,
8:49	and then a KUKA robot arm grabs them	and then a KUKA robot arm grabs them
8:50	and places them in one of 300 loading chutes	and places them in one of 300 loading chutes
8:53	made from 2.7 miles' worth of Hot Wheel tracks.	made from 2.7 miles' worth of Hot Wheel tracks.
8:57	You could load Dom directly with the arm,	You could load Dom directly with the arm,

8:59	but this is faster. He doesn't have to wait.	but this is faster. He doesn't have to wait.
9:01	He just comes to the docking station and the lower platform slides over	He just comes to the docking station and the lower platform slides over
9:05	so the bottom layer of 300 dominoes gets loaded up all at once.	so the bottom layer of 300 dominoes gets loaded up all at once.
9:08	We had a backup loading system using a tray	We had a backup loading system using a tray
9:10	just in case the robot arm wasn't working.	just in case the robot arm wasn't working.
9:13	Besides the Hot Wheels tracks, there's a ton of 3D printed parts	Besides the Hot Wheels tracks, there's a ton of 3D printed parts
9:16	throughout the build that we either printed ourselves,	throughout the build that we either printed ourselves,
9:18	or if we were in a pinch, my friends at MatterHackers helped us out.	or if we were in a pinch, my friends at MatterHackers helped us out.
9:22	So that's the loader. Now how about the Dominator himself?	So that's the loader. Now how about the Dominator himself?
9:25	How does Dom know exactly where to go in the room to drop a domino?	How does Dom know exactly where to go in the room to drop a domino?
9:29	So we've pre-programmed the route for all 102,000 dominoes	So we've pre-programmed the route for all 102,000 dominoes
9:32	so the robot knows exactly where to go from the start.	so the robot knows exactly where to go from the start.
9:35	As we're driving around, we use these indoor GPS sensors	As we're driving around, we use these indoor GPS sensors
9:38	to track the position of the robot so it knows where it is.	to track the position of the robot so it knows where it is.
9:41	As we get closer to the place the dominoes need to drop,	As we get closer to the place the dominoes need to drop,
9:43	we use IR cameras that are tracking markers on the ground	we use IR cameras that are tracking markers on the ground
9:46	to make sure the robot lines up every time.	to make sure the robot lines up every time.
9:48	So the vision is that we could set Dom up,	So the vision is that we could set Dom up,
9:51	turn off the lights and leave, come back the next morning	turn off the lights and leave, come back the next morning
9:54	and you've got a field of dominoes set up.	and you've got a field of dominoes set up.
9:56	-He could work in the dark just fine. - [Mark] How about these super cool wheels?	-He could work in the dark just fine. - [Mark] How about these super cool wheels?
10:00	[Alex] These are omni wheels. They're awesome.	[Alex] These are omni wheels. They're awesome.
10:02	They let you translate it any direction.	They let you translate it any direction.
10:04	This is way better than your car,	This is way better than your car,
10:06	where if you need to move a little to the left or right,	where if you need to move a little to the left or right,
10:08	you have to make a five-point turn.	you have to make a five-point turn.

10:10	With these, you can move any direction	With these, you can move any direction
10:12	to adjust for small corrections in the placement.	to adjust for small corrections in the placement.
10:15	These wheels are powered by ClearPath brushless DC servo motors.	These wheels are powered by ClearPath brushless DC servo motors.
10:18	These are the gold standard in terms of speed and accuracy.	These are the gold standard in terms of speed and accuracy.
10:21	They allow us to slide the robot around by even a couple of millimeters,	They allow us to slide the robot around by even a couple of millimeters,
10:24	which is super useful when we're trying to make fine adjustments.	which is super useful when we're trying to make fine adjustments.
10:28	Now, obviously that's just the uppermost tip of the iceberg tip,	Now, obviously that's just the uppermost tip of the iceberg tip,
10:32	but Dom's brain is over 14,000 lines of code, all from scratch.	but Dom's brain is over 14,000 lines of code, all from scratch.
10:36	If you want to go into the technical details,	If you want to go into the technical details,
10:38	I'll put a link in the description where Alex put it in one spot.	I'll put a link in the description where Alex put it in one spot.
10:41	Back to finishing the second half.	Back to finishing the second half.
10:47	♪ Enchanté ♪	♪ Enchanté ♪
11:14	So the Dominator has been running just under 24 hours,	So the Dominator has been running just under 24 hours,
11:18	and this is a bit of a momentous tile here.	and this is a bit of a momentous tile here.
11:24	It's-a me, Mario!	It's-a me, Mario!
11:31	102,300 dominoes are on the floor successfully.	102,300 dominoes are on the floor successfully.
11:37	This is the last 300.	This is the last 300.
11:39	I think this is a moment.	I think this is a moment.
11:41	-[music stops] -You guys, get back over here.	-[music stops] -You guys, get back over here.
11:43	Josh and Alex, let's observe it.	Josh and Alex, let's observe it.
11:46	Don't blow this, Dom. You got it, buddy. Finish strong.	Don't blow this, Dom. You got it, buddy. Finish strong.
12:00	We got it.	We got it.
12:01	Just over 24 hours to lay all the dominoes.	Just over 24 hours to lay all the dominoes.
12:05	Gentlemen.	Gentlemen.
12:07	With everything set up,	With everything set up,
12:09	the final critical step was to knock 'em down.	the final critical step was to knock 'em down.
12:11	You can see John loading the Mario pipes. They were gonna be our trigger mechanism.	You can see John loading the Mario pipes. They were gonna be our trigger mechanism.
12:15	The way that works is we've got four Hot Wheels cars	The way that works is we've got four Hot Wheels cars

12:17	glued to the bottom of this platform that rides in these parallel Hot Wheels tracks.	glued to the bottom of this platform that rides in these parallel Hot Wheels tracks.
12:22	Then slap a green sliding shell on top,	Then slap a green sliding shell on top,
12:24	and a motor at the end pulls the cart along the tracks with a string.	and a motor at the end pulls the cart along the tracks with a string.
12:27	The shell contacts the lever arm at each pipe like this,	The shell contacts the lever arm at each pipe like this,
12:30	which causes them to roll forward down the ramp,	which causes them to roll forward down the ramp,
12:33	hopefully into the dominoes.	hopefully into the dominoes.
12:34	-Okay, now I'm nervous. -[chuckling]	-Okay, now I'm nervous. -[chuckling]
12:37	[Mark] I wasn't nervous. Now I'm nervous. Okay.	[Mark] I wasn't nervous. Now I'm nervous. Okay.
12:40	Three, two, one, go.	Three, two, one, go.
12:49	-Yes! Two. -[Alex] Oh.	-Yes! Two. -[Alex] Oh.
12:51	Three.	Three.
12:53	Four.	Four.
12:55	Five.	Five.
13:03	Oh my gosh.	Oh my gosh.
13:16	Got it! Whoo!	Got it! Whoo!
13:19	Yes!	Yes!
13:20	-That is awesome. -Oh!	-That is awesome. -Oh!
13:23	Holy cow.	Holy cow.
13:25	Whoa.	Whoa.
13:28	Oh my gosh.	Oh my gosh.
13:31	In the end, Dom was 50 times faster than a skilled domino builder,	In the end, Dom was 50 times faster than a skilled domino builder,
13:35	and he got a world record to show for it.	and he got a world record to show for it.
13:37	Although to be fair to us organisms, it was in the non-human category.	Although to be fair to us organisms, it was in the non-human category.
13:41	At that point, after five years, we were so relieved to be done,	At that point, after five years, we were so relieved to be done,
13:44	we got carried away celebrating the champ	we got carried away celebrating the champ
13:47	with an appropriate send-off.	with an appropriate send-off.
13:54	-What do we do now? -A robot to clean it up.	-What do we do now? -A robot to clean it up.
13:56	Oh! There we go.	Oh! There we go.
13:59	A cleanup robot.	A cleanup robot.